Meshing/ CFD Pipeline

Monday, April 24, 2023 2:44 PM

Order of Files to Run for Meshing Surfaces:

1. **surface_prep.py**
- The purpose of the surface prep file is to clip the segmentation, using tube clipper function which is used to map the mesh
2. **centerlines_fixed.py or make_centerlines.py**
3. **map_info.py** or **map_info_bilateral.py**
- The purpose of the map_info.py is to map the attributes such as Dh, CSA, perimeter, etc. to the surface file. This where the centreline is also generated. Most importantly, the Taylor lengths are generated here which are of most importance
4. **make_mesh.py**
- Using the attributes generated in map_info.py and surface from surface_prep.py, a tetrahedral mesh is generated, the minimum edge length should be the minimum edge length found in Taylor length attribute (in m convert to mm), and the maximum should be found based on the diameter ratio
5. **mesh_quality.py**
- The purpose of this file is to check the quality of the mesh, but looking at size distribution, angles etc.
6. **Sending jobs to scinet**

https://gitlab.com/ramen_newdals/oldCFDPipeline

For pressure maps: you need map it to

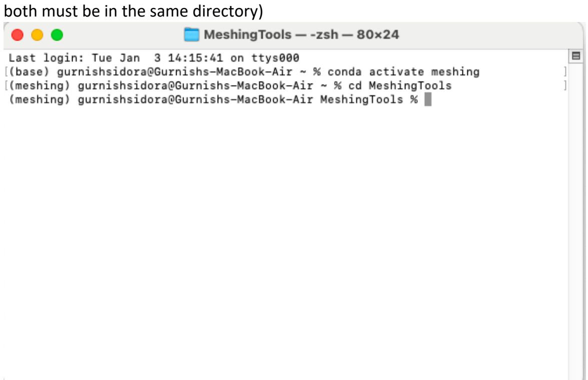
Based on Anna's Geometry Tools, set-up all the dependencies

- https://github.com/gsidora/geometry_tools

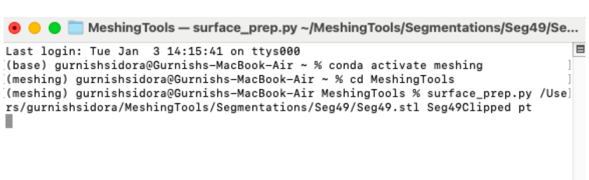
surface_prep.py

1. Open up the terminal and activate the meshing environment

2. Cd into the Directory containing the geometry tools and segmentations for the meshing (note: both must be in the same directory)

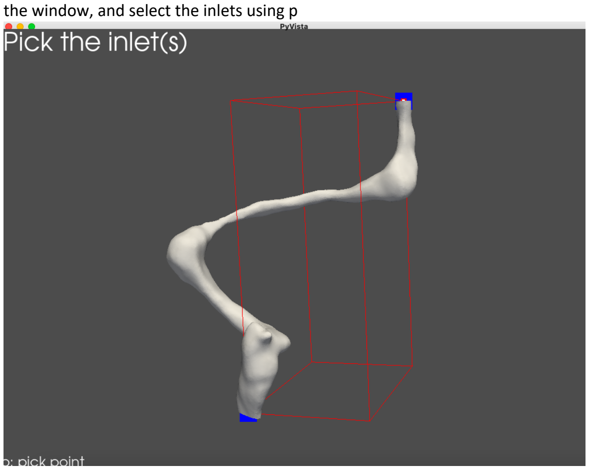


3. Run surface_prep.py in Geometry Tools to add flow extensions to the segmentation. Note: The segmentation must be saved in the same folder as the meshing tools
- Command to run surface_prep.py: surface_prep.py surf_file(stl) proj_dir surf_type(pt)



5. The following Pyvista window will pop-up and you will now be able to clip the boundaries. Press I where an interactive window will pop, place it at the ends, and press spacebar to clip

6. Once the boundaries have been clipped by selecting them interactively, as following, exit out of the window, and select the inlets using p

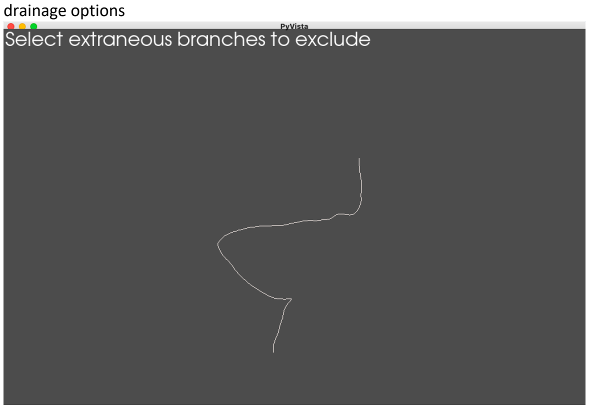


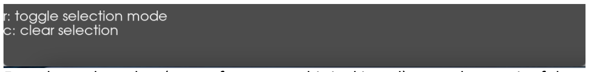
7. Inspect for unfilled holes, if present press q+r to re-clip the segment, otherwise press q to accept the surface.

map_info.py

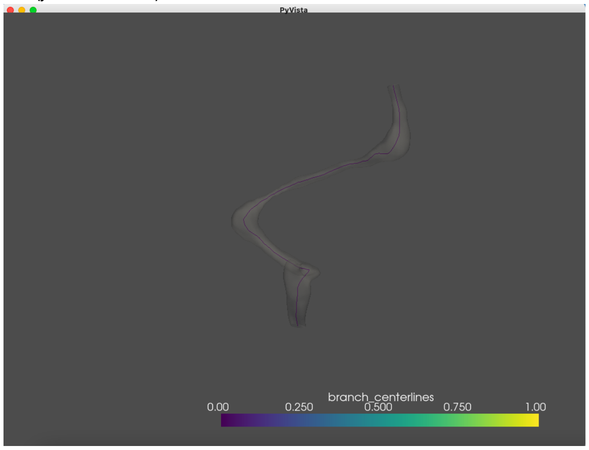
1. Run the following command in the meshing environment, and the following window will pop open

- map_info.py prep_dir(Directory containing surface_prep.py output) ss lab fen syl emissary condylar
- Everything after ss including ss are optional arguments requiring flow rates for the alternative drainage options





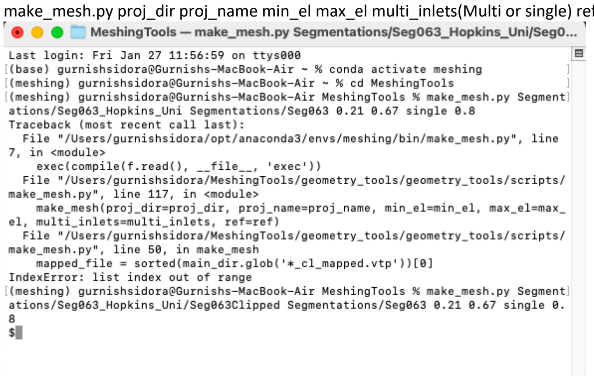
2. Extrude any branches (most often or not this is skipped), once done, exit of the window, and the following window will pop-up. Take a note of the centreline here, at certain parts it may appear odd (just bear with it)



3. The mapped files will be stored in the projdir as _mapped.vtp which will be required to determine the minimum and maximum edge lengths. The minimum edge length aligns with the minimum Taylor Length determined from map_info.py. The maximum EL should be average SSS/JV diameter/10. Keep in mind that the difference between Elmin and Elmax should not exceed 1.0mm.

make_mesh.py

1. Run the following command in the meshing environment and the following window will pop up.
   make_mesh.py proj_dir proj_name min_el max_el multi_inlets(Multi or single) ref (0.8)

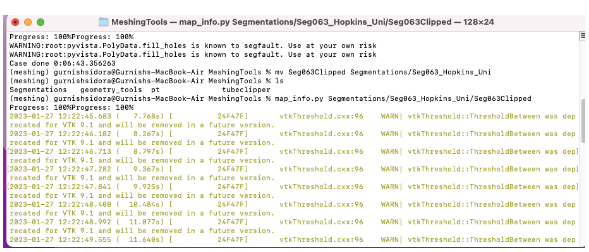


2. The following window will pop out, requiring to select inlet by pressing p.

3.

Notes:

- The Taylor lengths computed in the code are in metres, need to convert them to mm for the make_mesh.py code
- The input for emissary veins, and condylar veins are ratio outputs relative to the IJV, need to calculate them based on the diameters from the flow extensions (because its easier to calculate)
- Input the flow rates as 7 significant digits



Centreline is being problematic at the bulb --> difficult issue to fix

Not issues for mapping information, for visualization (in paper) it might be problematic)

- Only mapping things --> delta p on the centreline
- No impact on spectrograms -->
    - Brainstorm some ideas for how to address the centrelines --> figure it out on your own

Step size --> not a huge difference, wouldn't be able to Nyquist frequency, the higher the step size, the lower the Nyquist frequency, every 2 steps or 4 steps, step size should always be 1

Radius --> Does nothing for surface area

Select the entire region of the sigmoid sinus from the stenosis to the JV

SPIs to guide the results for the spectrograms --> but generate at the sigmoid sinus for all cases to show that might not always be the site of sound mechanisms